

Chemical Vapor Deposition Techniques

T.M. Besmann, D.P. Stinton, and
R.A. Lowden

Introduction

Chemical vapor deposition (CVD) is one of the few deposition processes in which the deposited phase is produced *in situ* via chemical reaction(s). Thus the vapor source for CVD can consist of high vapor pressure species at moderate temperatures and yet deposit very high-melting phases. For example, pure TiB_2 , which melts at 3225°C , can be produced at 900°C from TiCl_4 , BCl_3 , and H_2 .

Chemical vapor deposition and its variants such as low pressure CVD (LPCVD), plasma-assisted CVD (PACVD), and laser CVD (LCVD) have been active areas of research for many years. Recent review articles have contained extensive lists of the phases deposited by CVD, which include most of the metals and many carbides, nitrides, borides, silicides, and sulfides.¹⁻³ The techniques have found increased acceptance as commercial methods for the fabrication of films and coatings which are fundamental to the semiconductor device and the high-performance tool bit industries. They have been used to prepare multiphase-multilayer coatings, stand-alone bodies, and fiber-reinforced composites. As the demand increases for more complex and sophisticated materials, it is expected that CVD will play a still larger role.

In CVD a solid material is deposited from gaseous precursors onto a substrate. The substrate is typically heated to promote the deposition reaction and/or provide sufficient mobility of the adatoms to form the desired structure. Chemical vapor deposition was performed for the first time when early humans inadvertently coated cooking utensils with soot from the campfire. In this CVD process, hydrocarbons gener-

ated by the heated wood pyrolyzed on the utensil surface, depositing carbon.

One of the most important advantages of CVD is its high deposition rate, which can be greater than tens of micrometers per hour. It is substantially exceeded only by techniques such as plasma spraying. Its other advantages include control over stoichiometry, morphology, crystal structure, and orientation, which can be tailored by controlling deposition parameters. Since the technique does not require line-of-sight with the vapor source, coatings can be uniformly deposited over substantial contours. Chemical vapor deposition is also a relatively low-temperature process, as noted earlier, allowing the preparation of refractory materials at a small fraction of their melting temperature. Indeed, the low temperature together with the ease of purifying the reactants allow the deposition of exceptionally high-purity materials. Like ion plating and sputtering, however, appropriate parameters in PACVD can result in desired surface modification through the bombardment of the substrate by high energy species.⁴

The principal disadvantages of CVD include even the moderate heating of the substrate and the typical corrosiveness and toxicity of the process gases. The elevated temperatures needed to efficiently deposit films and coatings and obtain desired structures can exceed the limits of the substrate, causing phase changes or grain growth and diffusion of dopants in microelectronic components. The often corrosive reactant and product species in CVD can attack substrate materials, causing porosity, poor adherence, and contamination of the coating. Finally, CVD is largely an "equilibrium" process,

although it is common to produce amorphous films of certain materials. Thus metastable materials easily deposited by techniques such as sputtering may not be reproduced using CVD techniques.

CVD Systems

Chemical vapor deposition systems have several common components: (1) a vapor precursor feed system, (2) the CVD reactor (a furnace or similar system for heating the substrate), and (3) an effluent gas handling system (exhaust gases are often corrosive or toxic, or both). Vapor precursors are typically metal halides and, depending on the type of reaction and material to be deposited, they are flowed together with a reductant (H_2) or nonmetal source gas (e.g., oxidant). Some metals, however, do not form high vapor pressure halides, and therefore a number of metalorganic vapor sources have been developed, opening a new area of CVD. Deposition reactions include pyrolysis, reduction, oxidation, hydrolysis, disproportionation, or combinations of these, and may be catalyzed by the substrate. A common CVD reaction is the deposition of SiC from methyltrichlorosilane ($\text{CH}_3\text{Cl}_3\text{Si}$) in the presence of hydrogen.

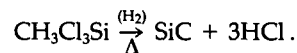


Figure 1 shows a commercial CVD apparatus.

Conventional CVD systems are classified as having either hot-wall or cold-wall reactors. Hot-wall reactors are heated chambers (furnaces) into which the substrate is placed for coating. As expected, the interior of the furnace also becomes coated, resulting in both maintenance problems and lower efficiency. In cold-wall reactors only the substrate is heated, either inductively or resistively, and although these reactors are more complex, they allow greater control over the deposition process.

Two variants of CVD which are receiving increased interest are PACVD and LCVD. In PACVD energy is imparted more directly to the reactant gases by exciting either those species or a carrier gas (e.g., argon), which in turn transfers energy to the reactant species. Radio frequency (rf) and microwave induction are the most common methods for generating plasmas. In these systems the substrates must still be heated—albeit often to a lower temperature than in conventional CVD—in order to have sufficient adatom mobility/reactivity to obtain the desired struc-

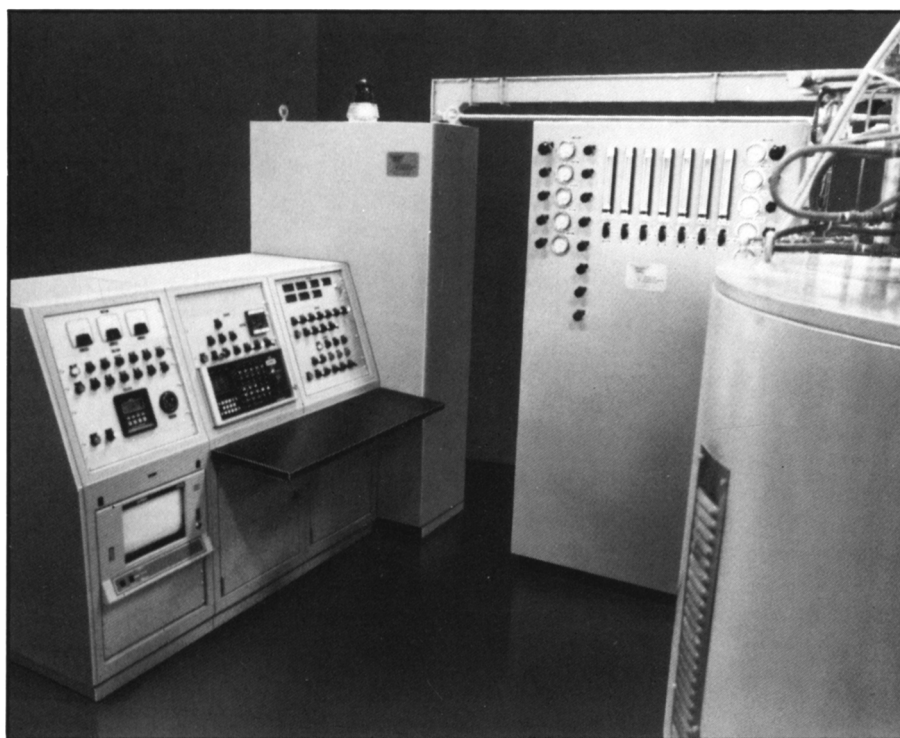


Figure 1. Photograph of a commercial CVD system (Multi-Arc Division, Andar Corp., St. Paul, MN).

ture/crystallinity and adherence. The principal advantages of PACVD are the use of lower substrate temperatures and a greater potential for producing metastable phases. The technique, however, increases the number of deposition parameters which must be controlled, is more cumbersome than conventional CVD, and requires greater capital investment.

Laser CVD utilizes laser light energy either to activate gaseous species (photolysis) or heat the substrate to promote CVD reactions. Thus in laser CVD the beam can be directed parallel to the substrate, close to the surface, or normal to the surface, depending on which technique is used. The special advantage of this method is the ability to "write" CVD patterns with the laser, restricting the deposition to selected submicrometer lines and areas—a particularly neat trick for microelectronic device fabrication. Photolysis also requires minimal heating of the substrate, as in PACVD. In addition, laser irradiation of the substrate heats only the near surface volume; thus the bulk of the substrate is not damaged. Disadvantages include limited throughput, the cost and problems associated with relatively high

power lasers, and required line-of-sight with the laser.^{1,5,6}

In this review of CVD we separate the technology into three application categories: (1) surface modification/coating, (2) composites by CVD infiltration, and (3) electronic materials. We briefly discuss the materials and techniques used to fabricate these products with a view toward the state of the art.

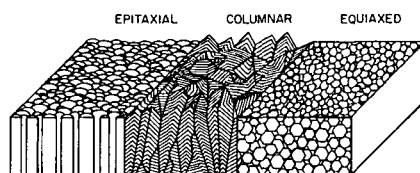


Figure 2. Schematic representation of structures deposited by CVD.

Surface Modification/Coating

Structural materials frequently possess the required mechanical properties for a specific application but lack the necessary surface characteristics (corro-

sion resistance, oxidation resistance, or wear resistance). It is often more economical to apply a thin coating possessing these desired properties to the substrate than to fabricate the entire component from a more costly or difficult-to-form material.

A major advantage of CVD processes for producing coatings is the ability to control the microstructure, which can vary widely depending on the deposition conditions (Figure 2).⁷⁻⁹ Of course, microstructural control is necessary to optimize the mechanical properties of a material. Epitaxial deposits result from conditions of low supersaturation (low reactant concentrations) and sufficiently rapid surface diffusion. Deposits consisting of randomly oriented fine grains near the substrate surface that grow into large columnar grains of preferred orientation are due to higher supersaturation and more limited diffusion. The most desirable coatings for mechanical strength and fracture toughness consist of very fine, equiaxed grains. This microstructure is typically produced at still higher supersaturations and under conditions which limit surface diffusion. The supply of reactants must be sufficient to rapidly renucleate new grains while the movement of atoms to preferred crystal lattice sites is restrained. Addition of trace quantities of impurities or catalysts may promote renucleation.

Coating adherence is a critical problem that must be addressed in every coating application. The coating's coefficient of thermal expansion must closely match that of the substrate to prevent cracking or spalling. Intermediate coatings with thermal expansion values between those of the substrate and surface layer are often necessary to minimize the residual stresses created at the interface. More recently, coatings have been produced in which the composition of the interlayer gradually varies from that of the substrate to that of the coating. This variation not only reduces the residual stresses but also avoids interface effects. Coating compositions must also be selected to avoid significant reaction at the coating-substrate interface. Reaction layers frequently create large residual stresses that result in cracking or spalling.

A recently developed method for increasing adherence is to increase the fracture toughness of the coating to lessen the chance of cracking. Fracture-toughened coatings can be produced by codeposition of more than one phase. Composite coatings exhibit increased toughness because cracks that propa-

gate in the coating interact with the included second phase and are blunted or deflected. For example, the codeposition of a SiC-TiSi₂ layer in place of SiC increased the fracture toughness of the coating by a factor of two.¹⁰

A major market for coatings is the cutting tool industry, where nearly 50% of the cobalt-bonded tungsten carbide tool inserts are coated with TiC, TiN, or TiC_xN_{1-x}.^{11,12} Coatings extend the life of these inserts by a factor of five because of their increased hardness and wear resistance, and by preventing reaction of the cobalt binder with the metal workpiece at the elevated temperatures of machining. The complementary properties of the carbide and nitride reduce sliding and crater wear, and further improve their performance. Unfortunately, titanium carbide and nitride are susceptible to oxidation at temperatures above ~800°C. Thus for tool bits used in high temperature, high speed operations, oxidation resistant Al₂O₃ coatings with high hardness are applied to the outer surface. At temperatures greater than 800°C, Al₂O₃ coated inserts have a life twice that of comparable TiC-coated inserts. Multiple layers of different materials are routinely used to combine the best properties of each layer. For example, TiC interlayers are frequently used between the substrate and TiN or Al₂O₃ coatings to improve adherence.

Hard coatings have also found application in erosion protection in gas turbine engines, rocket nozzles, and let-down valves in fossil energy systems. Erosion of metallic blades by abrasive particles in gas turbine engines significantly shortens the life of the blades and reduces efficiency. Titanium carbonitride coatings nearly eliminate erosive wear in this application and extend the lifetime by a factor of 70 or 80.¹³

The high temperature mechanical and physical properties of carbon fibers and carbon/carbon composites have led to their widespread use as refractory materials in aerospace heat shield and structural components, and more recently in turbine propulsion systems.¹⁴ These and other applications have been restricted, however, by the low oxidation resistance of carbonaceous materials at elevated temperatures. Efforts have expanded to develop protection systems for carbon materials which will allow long term exposure to high temperature oxidizing environments. Chemical vapor deposition processes have been adopted to produce theoretically dense single and multilayer coat-

ings that fulfill oxidation protection requirements.

A concept using SiC coatings has been successfully demonstrated on the structural components of reusable space vehicles which encounter re-entry temperatures greater than 1300°C. A coating of SiC on a carbon/carbon composite prevents carbon diffusion, and the subsequent deposition of compounds that form a glass layer upon oxidation seals the surface. The viscous glass outerlayer acts as the oxygen barrier and seals cracks that may form in the SiC coating from coating/composite thermal expansion mismatch. Doping of the silicon carbide coatings with boron, or the deposition of boride layers, further reduces the rate of oxidation.¹⁵⁻¹⁷

Freestanding shapes can also be fabricated by CVD techniques. Components are produced by depositing a relatively thick coating onto a mandrel and subsequently removing the mandrel. The dimensions of the component are controlled by the shape of the mandrel and the thickness of the applied coating. An application of this technique is the production of optically transparent windows for missile guidance systems. Chemically vapor-deposited polycrystalline ZnS or CdS, in the 8-12 μm range necessary to perform in this application, have the requisite hardness, fracture toughness, erosion resistance, and—most importantly—are transparent to electromagnetic radiation. The microstructure, mechanical properties and optical properties of these materials are optimized by control of the deposition temperature, reactant gas concentration, and growth rate. Freestanding BN shapes such as crucibles and cylinders are also fabricated by CVD. Boron nitride is typically applied to graphite shapes and, if properly designed, the graphite mandrel is easily removed.¹⁸

Fiber-Reinforced Composites

Fiber-reinforced composites are being fabricated using CVD to make up the matrix material. This process, called chemical vapor infiltration (CVI), is one of only a few processes capable of incorporating continuous ceramic fibers into a ceramic matrix without chemically, thermally, or mechanically damaging the relatively fragile reinforcing fibers. The high strength (~400 MPa) and exceptional fracture toughness (>10 MPa · m^{1/2}) of these composites, combined with their refractoriness and resistance to erosion, corrosion, and wear, make them ideal for numerous advanced high temperature structural

applications.

Commercial interest is focused on composites fabricated from continuous silicon carbide or alumino-silicate fibers and a matrix of SiC. Silicon carbide is typically deposited from methyltrichlorosilane following reaction 1. Deposition of stoichiometric SiC is promoted with this reactant because of the 1:1 gram-atomic ratio of silicon to carbon in the reactant. Process parameters vary from one fabricator to another but normally involve deposition temperatures of ~1200°C and furnace pressures of ~3 kPa. Composites can be prepared this way with densities approaching 90% of theoretical.

Current commercial CVI processes are isothermal and require reactant gases to diffuse into the freestanding fibrous preform. They are well suited for fabricating thin-walled composites of complex shapes (Figure 3).^{19,20} Densification, however, is inherently very slow because of the necessity for diffusion to carry the reactants into the preform and the gaseous reaction products back to the preform surface. In addition, the deposition rate is intentionally restrained by low temperature, reactant concentration, and pressure to prevent sealing the exterior surface of the preform before the interior can be densified. Despite these efforts, greater deposition of matrix material still occurs on fibers near the surface of the preform than occurs on fibers near the center, resulting in significant density gradients. To reduce the density gradients, the process is repeatedly interrupted for intermediate machining operations to remove material from the surface and open passageways to the interior of the preform. This cyclic process routinely requires three weeks to several months to complete. Despite these precautions, severe density gradients often occur in composites >3 mm thick. However, process economics can be achieved by using large furnaces that contain multiple preforms.

A different process for fabricating fiber-reinforced ceramic composites has been under development at Oak Ridge National Laboratory (ORNL) for several years. This process simultaneously utilizes thermal and pressure gradients to reduce the infiltration time from weeks to less than 24 h (Figure 4).²¹⁻²³ Fibrous preforms are retained within a graphite holder that contacts a water-cooled metal gas distributor, cooling the gas inlet and side surfaces of the substrate. The opposite end of the fibrous preform is exposed to the hot zone of the fur-

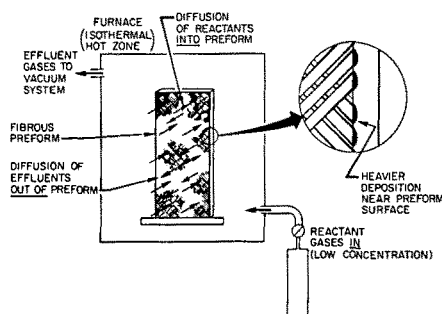


Figure 3. Commercial infiltration process utilizing isothermal hot zone.

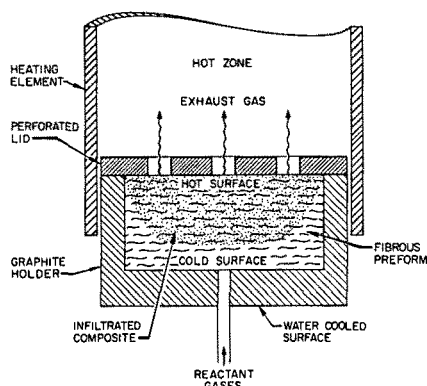


Figure 4. Schematic of the CVI process utilizing thermal and pressure gradients.

nance, creating a steep temperature gradient across the preform. The reactant gases are forced under pressure into the cooled side of the fibrous preform, but initially do not react because of the low temperature. The gases continue from the cooled portion of the preform into the hot portion, where the matrix begins to deposit on the fibers. Deposition of matrix material within the hot region of the preform increases the density and therefore the thermal conductivity of the preform. Thus the deposition zone moves progressively from the hotter regions toward the cooler regions. The process continues until reduced permeability of the densified composite prevents sufficient flow of reactant gases into the preform (a back-pressure of ~ 0.2 MPa).

The ORNL process is well suited for fabricating thick-walled composites of relatively simple shapes. Several advantages are inherent in this process, the primary one being increased efficiency and reduced infiltration time. Second,

the fabrication of preforms is simplified because fibrous materials are held in place by a graphite holder instead of requiring careful processing with binders that must be gently removed. Third, moderate pressures can be applied to the graphite holder to compress the preform and increase the fiber loading of the composite. Finally, composites with greatly increased thicknesses can be efficiently fabricated; specimens 25 mm thick have been produced with no significant density gradient between the surface and interior. Also, the flexibility of the process has been demonstrated by the fabrication of tubes with diameters up to 50 mm and wall thicknesses of about 10 mm.

Electronic Materials

Thin CVD films of insulators, dielectrics, semiconductors, and more recently conductors (metals and silicides) and magnetic materials, are extensively used in fabricating solid-state devices.²⁴ Film characteristics such as uniform coverage and high purity, and processing benefits such as low temperature, structural control, and deposition selectivity allow the production of unique, high quality thin films. At the same time, high throughput and low cost make the CVD of thin films economically attractive.

Vapor phase reactions can be adapted to deposit thin uniform coatings on the surfaces of devices containing complex geometries. This capability, combined with deposition selectivity and structural control, can be employed to overcome problems in processing very large-scale integrated (VLSI) fine-line devices. These high density devices may contain windows or interconnections of sub-micrometer dimensions and thicknesses. Other deposition techniques such as sputtering and evaporation are line-of-sight and are susceptible to shadowing and poor uniformity, making them unsuitable for this and similar applications. Nucleation and growth in CVD film formation are generally surface dependent. Thus coatings can be selectively deposited by surface treatment and area deactivation.

Substrate temperatures typically must be minimized so that previous processing, such as doping, is unaffected. Thus inorganic, organic, and metalorganic CVD sources that react at temperatures $<600^\circ\text{C}$ are required. The development of metalorganic compounds has specifically benefited the state of low temperature deposition and has broadened material selection. Pre-

cursors for CVD processes can usually be obtained in high purity form to allow films to be deposited with the low impurity levels needed in electronic devices.

Sophisticated reactor designs, automated mass flow and vapor source controllers, and feedback pressure control systems are used in the deposition of electronic materials. Low temperature reactors operating at atmospheric pressure are employed to deposit amorphous oxides and glasses for solid-state devices. Systems of various configurations include horizontal tube reactors, rotary vertical flow reactors, and continuous process reactors for large-scale production. Higher temperature furnace systems, designed to synthesize complex layer structures for microelectronic devices and to deposit polycrystalline films from halides or hydrides, are generally hot- or cold-walled silica glass tube designs. These systems require carrier gases to aid in delivery of reactants. Reduced pressure reactors avoid the use of the excess gases, decreasing particle contamination and improving uniformity; thus LPCVD techniques have been emphasized in recent years.

Plasma-assisted CVD has been widely employed in fabricating microelectronic components, largely due to the lower substrate temperatures it allows. A typical reactor, shown in Figure 5, rotates the substrate support to improve deposition uniformity. This technology is relatively mature for microdevices; however, it has yet to be satisfactorily scaled to the production of components larger than a silicon wafer. Examples are photovoltaic modules, xerographic photoreceptors, large area displays, and other thin film transistor-based technologies.²⁵

Plasma-assisted CVD has also been used extensively to prepare hard, diamondlike,²⁶⁻²⁸ and more recently diamond films,²⁹ from hydrocarbon plus hydrogen precursors. Diamondlike films are hard, electrically insulating, chemically inert, and optically transparent, although they lack long-range order and contain large amounts of hydrogen or inert gas. Because of their chemical inertness, diamondlike carbon coatings function as protective coatings against corrosion and as a protective or insulating coating for electronic components. This unique combination of properties also makes diamondlike carbon an ideal candidate for optical coatings, particularly since the refractive index can be controlled by deposition conditions.

PLASMA ACTIVATED CHEMICAL VAPOR DEPOSITION

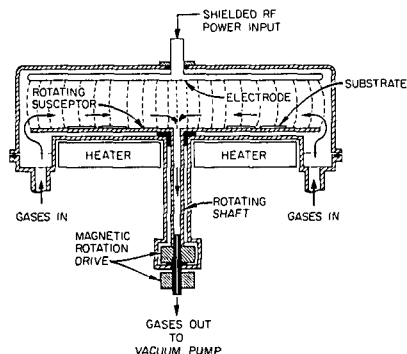


Figure 5. Schematic of a PACVD system.

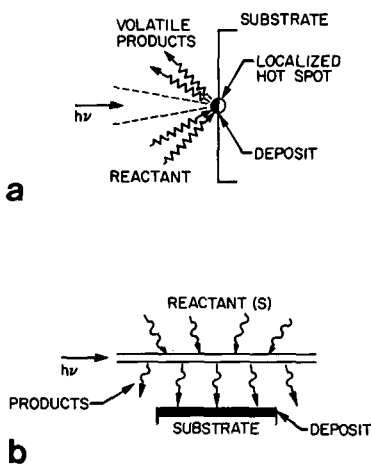


Figure 6. Laser CVD configurations illustrating (a) laser-heated surface upon which CVD reactions can occur and (b) photolysis.

The development many years ago of diamond films by Russian and Japanese scientists has recently been resurrected in a flurry of activity primarily in Japan and the United States. Diamond has a wide range of desirable properties and the ability to produce large area continuous films allows it to be used as thermally conducting but electrically insulating substrates, as doped semiconductors, as lenses for visible and infrared optics, and also in applications for hard coatings. Diamond film preparation has been accomplished most effectively by PACVD, and it appears to require the presence of atomic hydrogen to promote formation of the diamond structure over that of the more thermodynamically stable graphite.²⁹

As noted in the discussion of CVD systems, laser CVD is performed by either photolysis or laser heating of the substrate. The configurations for these reactors is shown schematically in Figure 6. The direct deposition of lines by laser CVD, however, is limited by the time required to write a pattern on a large number of wafers compared, for example, to wide area metallization. There is the potential that wide area laser beams may achieve sufficiently high deposition rates, possibly with high repetition rate excimer lasers. Films of aluminum, chromium, tungsten, and molybdenum have been deposited at rates up to 250 nm/min over 20 cm² areas.⁵ Excellent step coverage is also obtained using photolytic laser CVD as a result of the planar source of depositing species.

Several insulator films have been deposited by laser CVD, including zinc oxide, silicon nitride, and aluminum nitride. The PACVD of insulating films has been limited because of problems of plasma interaction with the chamber walls, causing increased impurity levels in films and radiation damage. Laser CVD has successfully avoided these problems.

Conclusions

Chemical vapor deposition is well suited for preparing many advanced products, including single-phase and composite materials, coatings, and films. Significant success in the manufacture of electronic semiconductors, beginning in the early 1970s, accelerated the development of CVD processes for the production of other materials. Hard and wear-resistant materials (TiC, TiN, TiB₂, Al₂O₃, and diamond and diamondlike carbon) have found numerous and growing applications such as coatings for cutting tool inserts, knives, and metallic turbine blades. The high strength and exceptional fracture toughness of fiber-reinforced ceramic matrix composites fabricated by CVI make them applicable for many high temperature structural applications. Multilayer CVD coatings of SiC/glass or SiC/oxide have proved effective for protecting carbon/carbon composites from high temperature oxidation. Application of PACVD has continued in the electronics industry for the manufacture of semiconductor devices, protective coatings for electronic circuits, and solar cells. Laser-assisted CVD is finding increased utility in component preparation, and, should its technological limitations be overcome, could play a major role in

producing radiation damage free, high density, complex integrated circuits and other electronic components.

Acknowledgments

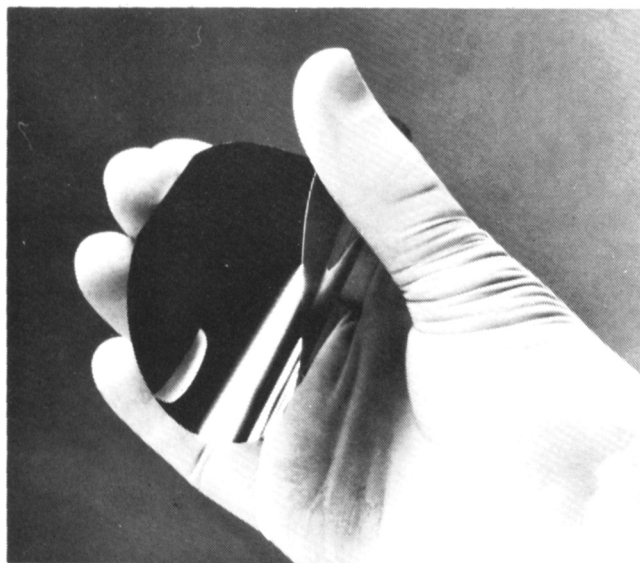
The authors would like to acknowledge the helpful reviews by E.L. Long Jr. and L.F. Allard. They would also like to thank C.A. Valentine for manuscript preparation and H.R. Livesey for the graphics.

Research sponsored by the Office of Energy Utilization Research, Energy Conversion and Utilization Technologies (ECUT) Program and the AR&TD Fossil Energy Materials Program, U. S. Department of Energy under contract DE-AC05-84OR21400 with Martin Marietta Energy Systems, Inc.

References

1. F.A. Houle, *Appl. Phys. A* **41** (1986) p. 315.
2. D.P. Stinton, T.M. Besmann, and R.A. Lowden, *Ceram. Bull.* **67** (2) (1988) p. 350.
3. M.L. Green and R.A. Levy, *J. Metals* (June 1985) p. 63.
4. R.F. Bunshah, in *Deposition Technologies for Films and Coatings*, edited by R.F. Bunshah et al. (Noyes Publications, Park Ridge, NJ, 1982) p. 1-18.
5. R. Solanki, C.A. Moore, and G.J. Collins, *Solid State Tech.* (June 1985) p. 220.
6. R.M. Osgood and H.H. Gilgen, *Ann. Rev. Mater. Sci.* **15** (1985) p. 549.
7. R.F. Bunshah and D.M. Mattox, *Phys. Today* (May 1980) p. 50-55.
8. J.A. Thornton, *Ann. Rev. Mater. Sci.* **7** (1977) p. 239-260.
9. W.A. Bryant, *J. Mater. Sci.* **12** (1977) p. 1285-1306.
10. D.P. Stinton and W.J. Lackey, *Ceram. Eng. & Sci. Proc.* **6** (7-8) (1985) p. 707-713.
11. R. Bonetti, *Met. Prog.* (June 1981).
12. H.E. Hintermann, *Thin Solid Films* **84** (1981) p. 215-243.
13. C.L. Yawsend and G.F. Wakefield, in *Proceedings of the Fourth International Conference on Chemical Vapor Deposition*, edited by G.G. Wakefield and J.M. Blocher Jr. (Electrochemical Society, Princeton, NJ, 1973) p. 577-587.
14. E. Fitzer, *Carbon* **25** (2) (1987) p. 163-190.
15. P. Ehrburger, P. Baranne, and J. Lahaye, *Carbon* **24** (4) (1986) p. 495-499.
16. D.W. McKee, C.L. Spiro, E.J. Lomby, *Carbon* **22** (6) (1984) p. 507-511.
17. D.W. McKee, *Carbon* **26** (6) (1986).
18. N.J. Archer, in *High Temperature Chemistry in Inorganic and Ceramic Materials*, edited by F.P. Glasser and P.E. Potter (The Chemical Society, London, 1977) p. 167-180.
19. E. Fitzer and R. Gadow, *Am. Ceram. Soc. Bull.* **65** (2) (1986) p. 326-335.
20. P.J. Lamicq, G.A. Bernhart, M.M. Dauchier, and J.G. Mace, *ibid.* p. 336-338.
21. D.P. Stinton, A.J. Caputo, and R.A. Lowden, *ibid.* p. 347-350.
22. A.J. Caputo, D.P. Stinton, R.A. Lowden, and T.M. Besmann, *Am. Ceram. Soc. Bull.* **66** (2) (1987) p. 368-372.

"ULTRA THIN"



**One-mil thick
silicon wafers!**

**We should say
one-mil THIN
wafers!**

**Two-, three- and four-inch diameter
double-sided polished wafers with:**

- less than 3 microns surface flatness
- less than 2.5 microns taper

Also available in 1-, 1½-, 2¼-inch diameter and other thicknesses. Thickness variation on wafer lot can be maintained to $\pm .0001$ ".

VSI makes the most dimensionally stable thin wafers. In addition, our throughput capabilities exceed 3,000 thin wafers/week. All processing, from crystal growth to polishing, is done on VSI premises.

For your
thin wafer needs,
call the leader!



VIRGINIA SEMICONDUCTOR, INC.

1501 Powhatan Street, Fredericksburg, VA 22401

Phone (703) 373-2900

Telex 9102506565 • Fax (703) 371-0371

CVD Techniques

23. A.J. Caputo and W.J. Lackey, *Ceram. Eng. & Sci. Proc.* 5 (7-8) (1984) p. 654-667.
24. A. Sherman, *Chemical Vapor Deposition for Microelectronics*, (Noyes Publications, Park Ridge, NJ, 1987).
25. J. Mort and F. Jansen, *Plasma Deposited Thin Films*, (CRC Press, Boca Raton, FL, 1986).
26. D. Koppers, in *Proceedings of the Seventh International Conference on Chemical Vapor Deposition*, edited by T.O. Sedgwick and H. Lydtin, (Electrochemical Society, Princeton, NJ, 1979) p. 159-175.
27. T.D. Bonifield, in *Deposition Technologies for Films and Coatings*, edited by R.F. Bunshah et al. (Noyes Publications, Park Ridge, NJ, 1982) p. 365-384.
28. J.C. Angus, P. Koidl, and S. Domitz, in *Plasma Deposited Thin Films*, edited by J. Mort and F. Jansen, (CRC Press Inc., Boca Raton, FL, 1986) p. 89-128.
29. M. Frenklach and K.E. Spear, *J. Mater. Res.* 3 (1) (1988) p. 133. □

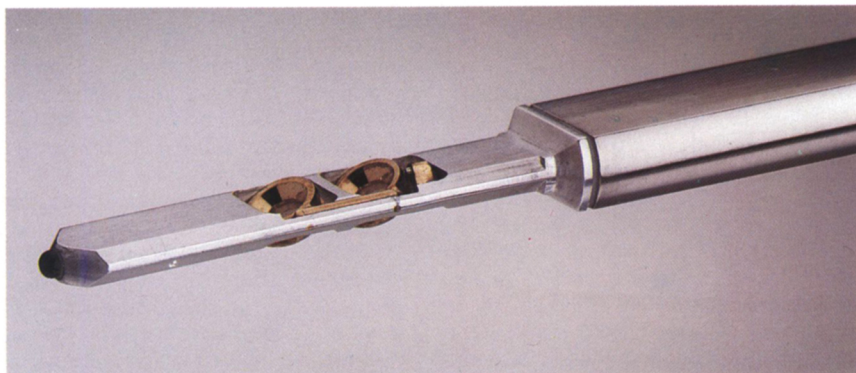
In the next issue...

Guest Editor Russell Messier continues the focus on Deposition Processes with articles on:

- ☐ Evaporation Processes
- ☐ Ion Beam Deposition, Film Modification and Synthesis
- ☐ Sputter Deposition Processes
- ☐ Plasma-Assisted Chemical Vapor Deposition
- ☐ Plasma-Sprayed Deposition Processes



HOLDERS and Conversions for TEMs



Twin specimen, double tilt holder

Gatan produced its first double tilt specimen holder in 1960 at a time when commercial TEMs offered only single tilt holders for stereo imaging. Since then we have made all types of specimen holder for almost every make and model of TEM.

Even today specialized specimen holders and stages remain a major part of our business and Gatan works closely with the leading TEM manufacturers to extend the versatility of electron microscopes through innovative holder design.

The sophistication of today's specimen holders inevitably makes them more delicate than simple single tilt types. For this reason, Gatan maintains excellent repair facilities capable of

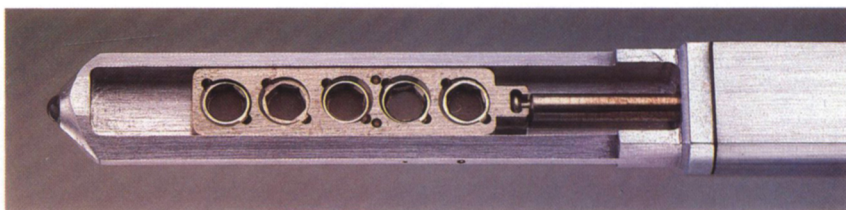
short turn around times so that should your Gatan holder be damaged, you will be without it for as short a time as is possible.

Gatan's designers are all experienced microscopists and understand the problems of specimen loading and manipulation. Our innovations include the Hexring™ specimen clamping system, the

Clipping™ cryotransfer clamping system, the robust, reliable toggle tilt mechanism, the bubble free liquid nitrogen dewar and the antidrift device (US Patent 4,703,181).

Two of our recent products are shown here but our complete line includes:

- *X-ray analytical (Be)
- Bulk analytical
- 360° Continuous Rotation (Be)
- Ultra high tilt analytical
- Twin specimen double tilt (Be)
- Multiple specimen single tilt (Be)
- *Vacuum transfer (Be)
- *Heating
- *Cooling (LN₂ and LHe) (Be)
- Cryotransfer system
- Micropositioning anticontaminator
- UHV stages (Philips TEMs)
- Controlled atmosphere cells
- (*Double and single tilt available)



Single tilt, multiple specimen holder

For further information on specimen holders, TV systems, EELS, PEELS™ and TEM specimen preparation equipment, please call our main sales and service offices at:

Gatan Inc., Pleasanton, CA
Tel. (415) 463-0200

Gatan Inc., Warrendale, PA
Tel. (412) 776-5260

Gatan GmbH, München, FRG
Tel. (089) 352374

Gatan Ltd., High Wycombe, UK
Tel. (0494) 26211

Please visit booth no. 309 at the MRS Show in Boston,
November 29-December 1, 1988.